Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	15 February 2026
Team ID	LTVIP2026TMIDS79776
Project Name	Cafeteria Menu Display
Maximum Marks	4 Marks

Technical Architecture:

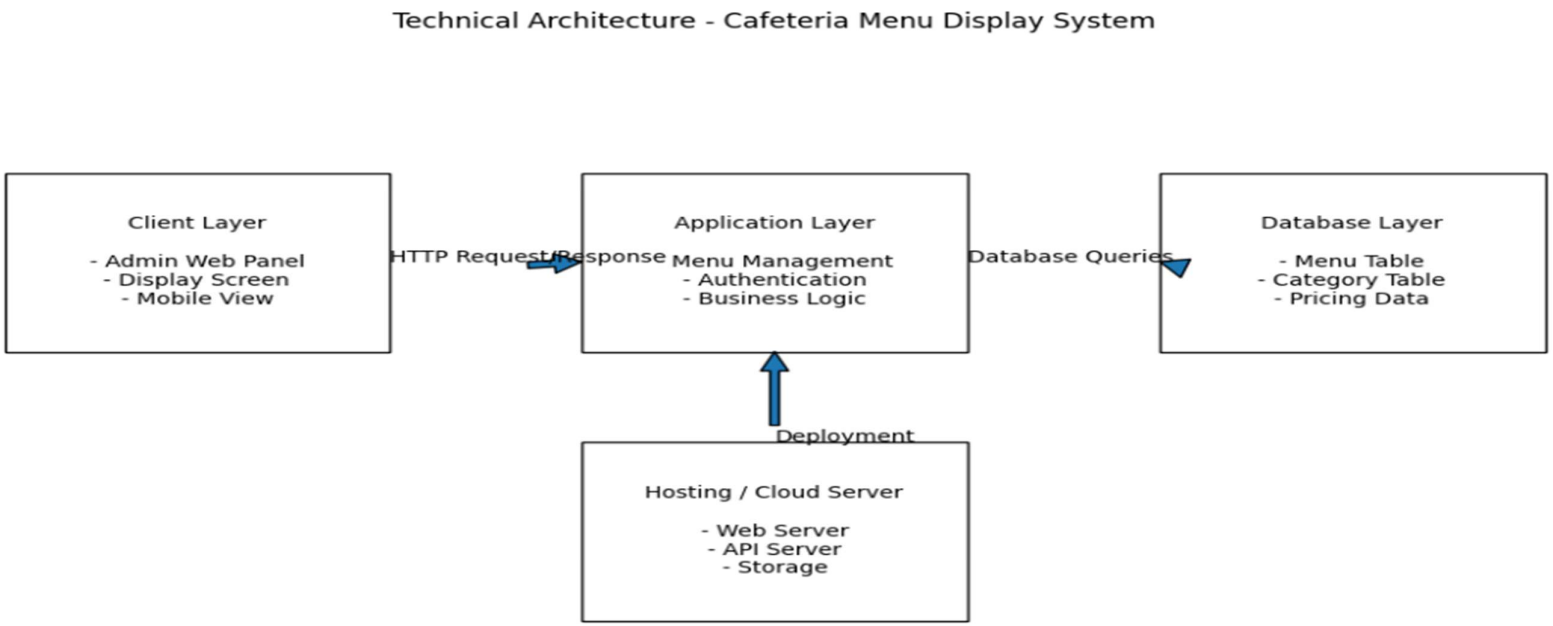

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How cafeteria staff and customers interact with the system.	HTML, CSS, JavaScript
2	Application Logic-1	Automated fetching and rendering of menu items on display.	JavaScript Fetch API
3	Application Logic-2	Admin authentication and menu management operations.	Python Flask(Backend)
4	Application Logic-3	Auto-refresh functionality every 5 minutes.	JavaScript setInterval
5	Database	Core tables for storing users, categories, and menu items.	MySQL Relational Database
6	Local Server	Centralized hosting of application and database.	Localhost / Flask Development Serve
7	Infrastructure	Application development and deployment environment.	VS Code, XAMPP, Git

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Scripting and UI frameworks utilized by the platform.	Flask, Bootstrap, jQuery
2	Security Implementations	Access controls for admin panel and sensitive menu data.	Session Management, Password Protection
3	Scalable Architecture	3-tier architecture separating UI, Logic, and Data layers.	Frontend (HTML/CSS/JS) + Backend (Flask) + Database (MySQL)
4	Availability	Accessibility of the cafeteria menu system.	Local Network / Localhost
5	Performance	Efficiency of real-time menu updates and auto-refresh.	Asynchronous JavaScript, Optimized Database Queries